

COURSE OUTLINE



Basic Information:		
Course Code: CSE 312		
Course Title: Database Management System Lab		
Credits: 1.5	Weekly: 3 Hours	Total Week: 14

Prerequisite:

Data Structure (CSE134)

Course Content (from syllabus):

Introduction to Database Systems and SQL Tools, Database and Table Creation with Constraints, Basic SQL Operations (SELECT, INSERT, UPDATE, DELETE), Advanced SQL Queries (Filtering, Grouping, Sorting), SQL Built-in Functions and Data Manipulation, Schema Design and Data Normalization (1NF to 3NF), Subqueries and Nested Queries, Joins and Set Operations, Views, Indexing, and Performance Optimization, Transaction Control and Concurrency Management, Database Security (User Roles, Privileges, Access Control), Distributed Database Concepts and Implementation, Introduction to NoSQL Databases (MongoDB) and Big Data Tools (Hadoop), Mini Project on Database Application Design, Project Presentation and Final Evaluation

Course Description/Rational:

This course is designed to provide hands-on experience in the design, implementation, and management of relational databases using structured query language (SQL) and modern database management systems. Through a series of structured lab exercises and projects, students will develop practical skills in database schema design, data manipulation, query optimization, and indexing techniques. The course also introduces students to advanced topics such as database security, distributed databases, and emerging database technologies (e.g., NoSQL, Hadoop). Emphasis is placed on applying theoretical knowledge to solve real-world problems using industry-standard tools and environments. This lab complements the theoretical aspects of database management systems by reinforcing core concepts and enhancing students' abilities to build efficient, secure, and scalable database solutions.

Course Objective:

1. Develop hands-on skills in creating, manipulating, and querying relational databases using SQL, including data definition and data manipulation commands.
2. Apply normalization principles and relational modeling techniques to design efficient, well-structured database schemas.
3. Implement and optimize relational databases using advanced SQL queries, indexing methods, and query optimization strategies to enhance real-world database performance.
4. Use modern database tools and technologies to simulate and analyze real-time data management problems, exploring security mechanisms and distributed architectures while fostering critical thinking and problem-solving skills.

Course Outcomes (COs) with Mappings:

After completion of this course, students will be able to:

COs	CO Statements	POs	Learning Domains	Knowledge Profile	Complex Engineering Problem	Complex Engineering Activities
CO1	Implement and optimize relational databases using basic to advanced SQL queries, indexing techniques, and query optimization strategies.	PO3	C3 C5 A2 P1 P2	K4 K5 K6	EP1, EP4	
CO2	Analyze and apply security techniques, distributed database architectures, and emerging tools in database management to address real-world challenges.	PO5	C3 C4 A2 P2 P3	K4 K6 K8	EP2	
CO3	Perform effectively as an individual or a member or a leader of diverse teams through proper documentation and initialization of project work.	PO9	A1 A2 A4 P3			
CO4	Create a project by explaining complex computer engineering activities with the computer engineering community by performing effective communication through demonstration and presentation	PO10	P2 A2 A4			EA1 EA2

Bloom's**Taxonomy****Cognitive Domain**

C2: Understand

C3: Apply

C4: Analyze

C5: Evaluate

C6: Create

Psychomotor Domain

P2: Manipulating

P3: Precising

Affective Domain

A1: Receiving/ Attending

A2: Responding

A3: Valuing,

A4: Organizing

Knowledge Profile

K3: Engineering

Fundamentals

K4: Specialist Knowledge

K5: Engineering Design

K6: Technology

K8: Emerging and Advanced Knowledge

CEP Attributes

EP1: Depth of Knowledge

EP2: Range of Conflicting Requirements

EP4: Familiarity of issues

Justification of CO-PO Mapping:

COs	POs	Justification of Mapping
CO1	PO3	CO1 is mapped to PO3 because students design and implement relational databases using SQL, indexing, and query optimization, addressing solution design for complex engineering problems.
CO2	PO5	CO2 maps to PO5 as students apply modern database tools (e.g., MongoDB, Hadoop) and implement real-world database solutions involving security and distributed systems.

CO3	PO9	CO3 maps to PO9 by teamwork, leadership, and documentation skills, enabling students to function effectively in individual and team-based project environments.
CO4	PO10	CO4 is aligned with PO10 because students demonstrate and present database projects, developing effective oral and written communication within engineering contexts.

Justification of Knowledge Profile (KP) Mapping:

COs	Knowledge Profile (K)	Justification of Mapping
CO1	K4 K5 K6	CO1 is mapped to K4 (Specialist Knowledge) as students gain in-depth understanding of relational modeling, indexing, and SQL optimization. It aligns with K5 (Design Knowledge) because students apply structured techniques to design efficient database schemas and queries. K6 (Engineering Practice Knowledge) is included as students carry out lab-based implementations using modern DBMS tools, developing practical skills in database tuning and deployment.
CO2	K4 K6 K8	CO2 is mapped to K4 (Specialist Knowledge) as it requires students to understand complex topics such as database security and distributed systems. It includes K6 (Engineering Practice Knowledge) because students implement secure and distributed solutions through hands-on lab exercises. K8 (Research and Development Knowledge) is addressed as students explore NoSQL databases, Big Data tools, and current trends, fostering innovation and critical analysis of evolving technologies.

Justification of Complex Engineering Problem (CEP) Attributes Mapping:

COs	CEP Attributes (EP)	Justification of Mapping
CO1	EP1, EP4	CO1 maps to EP1 and EP4 because it involves solving complex engineering problems related to designing, implementing, and optimizing relational databases using advanced SQL and indexing techniques, requiring critical evaluation and performance tuning skills.
CO2	EP2	CO2 maps to EP2 as it addresses complex engineering problems involving security implementation, distributed database architectures, and application of emerging tools like NoSQL and Big Data frameworks in real-world scenarios.

Justification of Engineering Activities (EAs) Mapping:

COs	CEP Activities (EA)	Justification of Mapping
CO4	EA1, EA2	CO4 maps to PO10 as it involves effective communication skills required to present complex engineering projects clearly and professionally. EA1 corresponds to the preparation and planning of communication materials, while EA2 relates to delivering oral and visual presentations to convey technical information effectively.

Mapping of Assessment Tools with Course Outcomes (COs):

Assessment Tools	Course Outcomes (COs)	Explanation
Lab Project based on CEP and along with Report	CO1, CO2, CO3	This assessment evaluates students' ability to apply database management concepts and techniques in practical, real-world scenarios. It requires designing, implementing, and documenting a complete database system. The project report further assesses students' skills in collaboration, communication, innovation, and addressing ethical issues within database management.

Course Delivery Plan/Lesson Delivery Plan:

Weeks	Experiment Name with Details	COs	Activities	Tools used	Assessment Strategies
1	Introduction to DBMS, Types and SQL		Review the basic syntax and features of SQL. Practice SQL using simple exercises.	Oracle, MySQL, SQL Server	N/A
2	Introduction to ER Modeling and related SQL		Learn the basic concepts ER Model and basic of CRUD operations using SQL Queries	Oracle, MySQL, SQL Server	Lab Performance
3	Relational Algebra and SQL Queries		Learn how to write equivalent SQL Queries from Relational Algebra	Oracle, MySQL, SQL Server	Lab Performance
4	Normalization Forms with SQL		Learn the principles and applications of Normalization Forms with SQL.	Oracle, MySQL, SQL Server	Lab Performance
5	DDL & DML		Learn the principles and applications of DDL and DML	Oracle, MySQL, SQL Server	Lab Performance
6	CRUD Operations		Learn the principles and applications of CRUD Operations	Oracle, MySQL, SQL Server	Lab Performance
7	Joining Queries		Learn the principles and applications of Joining Queries	Oracle, MySQL, SQL Server	Lab Performance
8	Storage, Indexing Techniques and Transaction Processing & Security Issues		Learn the principles and applications Storage, Index Techniques and Transaction Processing with Log Table and Trigger	Oracle, MySQL, SQL Server	Lab Performance
9	Introduction to No-Sql Database & Types		Learn about Types of NoSQL databases: key-value, column-family, graph; Use cases and applications;	N/A	Lab Report
10	Database administration tasks and responsibilities, Backup and recovery		Learn about LOG, Trigger, Performance tuning and optimization, Monitoring and troubleshooting	Oracle, MySQL, SQL Server	Lab Report

Weeks	Experiment Name with Details	COs	Activities	Tools used	Assessment Strategies

PO Validation via Assessment:

Assessment Type	Mapped COs	Mapped POs	Justification
Lab Project based on CEP and along with Report	CO1, CO2, CO3	PO1, PO3, PO5	This assessment type validates students' ability to apply core database management skills (CO1), design and implement solutions (CO2), and address ethical, social, and professional issues in their project (CO3). PO1 (Engineering knowledge) is demonstrated through problem-solving with database concepts, PO3 (Design and development) is evident in the database system design and implementation, and PO5 (Modern tool usage) is reflected in the integration of data management tools and methodologies in the project.

Assessment Pattern:

Assessment Tools					Marks (Total=100)
Attendance					10
Lab Performance Test					25
Lab Report					25
Lab Project based on CEP along with Report	CO-wise Marks				
	CO1	CO2	CO3	CO4	40
	18	12	5	5	
Total Marks					100

Learning Materials:**Textbook/Recommended Readings:**

SL	Book Name	Author(s)	Publishers	Edition and year
1	Database System Concepts	Abraham Silberschatz, Henry F. Korth, S. Sudarshan	McGraw-Hill Education	7th Edition, 2019
2	Database Management Systems	Raghu Ramakrishnan, Johannes Gehrke	McGraw-Hill Education	4th Edition, 2018

Reference Books/Supplementary Readings:

SL	Book Name	Author(s)	Publishers	Edition and year
1	Database Design for Mere Mortals	Michael J. Hernandez	Addison-Wesley Professional	3rd Edition, 2013
2	SQL Queries for Mere Mortals	John L. Viescas, Michael J. Hernandez	Addison-Wesley Professional	3rd Edition, 2007
3	SQL Cookbook	Anthony Molinaro	O'Reilly Media	1st Edition, 2005

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Signature


